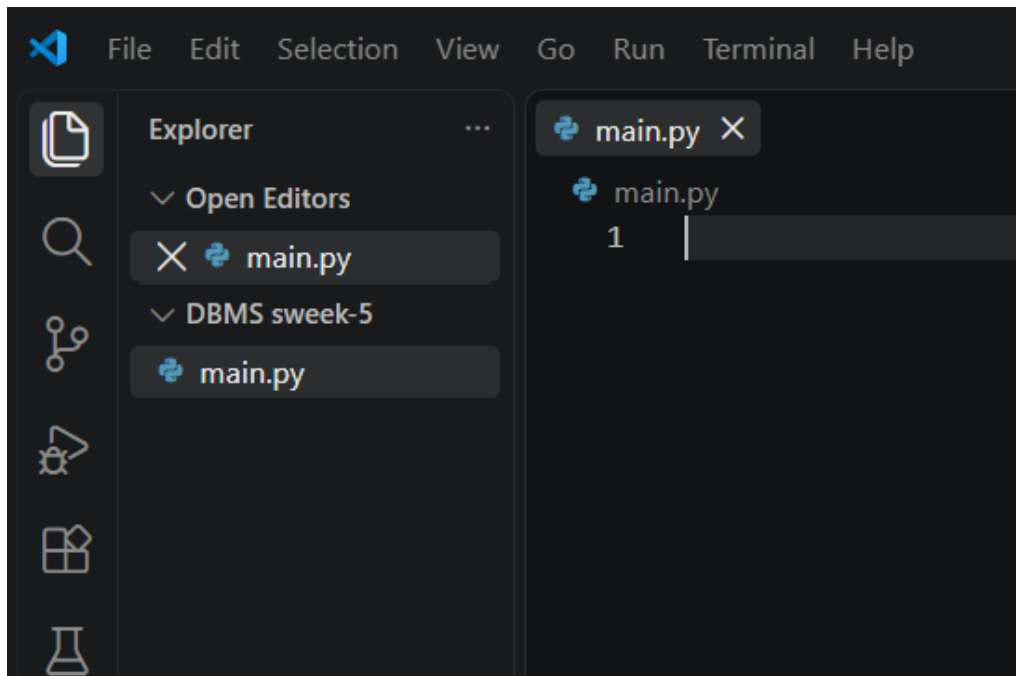


## Skill week - 5

---

### Step 1: Created Main.py file



### Step 2: Python version

```
PS C:\Users\saich\OneDrive\Desktop\HA\skill week5> python --version
Python 3.14.4
PS C:\Users\saich\OneDrive\Desktop\HA\skill week5>
```

### Step 3: Checked python PIP

```
PS C:\Users\saich\OneDrive\Desktop\HA\skill week5> python --version
Python 3.14.4
PS C:\Users\saich\OneDrive\Desktop\HA\skill week5> python -m pip --version
pip 26.2.1 from C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\pip (python 3.14)
PS C:\Users\saich\OneDrive\Desktop\HA\skill week5> █
```

### Step 4: Installed FastApi and Uvicorn

```
PS C:\Users\saich\OneDrive\Desktop\HA\skill week5> python --version
Python 3.14.4
PS C:\Users\saich\OneDrive\Desktop\HA\skill week5> python -m pip --version
pip 26.2.1 from C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\pip (python 3.14)
PS C:\Users\saich\OneDrive\Desktop\HA\skill week5> python -m pip install fastapi uvicorn
Requirement already satisfied: fastapi in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (0.141.1)
Requirement already satisfied: uvicorn in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (0.52.4)
Requirement already satisfied: starlette>=0.46.0 in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from fastapi) (1.6.0)
Requirement already satisfied: pydantic>=2.9.0 in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from fastapi) (2.13.4)
Requirement already satisfied: typing-extensions>=4.8.0 in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from fastapi) (4.16.0)
Requirement already satisfied: typing-inspection>=0.4.2 in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from fastapi) (0.4.4)
Requirement already satisfied: annotated-doc>=0.0.2 in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from fastapi) (0.0.5)
Requirement already satisfied: click>=7.0 in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from uvicorn) (8.4.2)
Requirement already satisfied: h11>=0.8 in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from uvicorn) (0.16.0)
Requirement already satisfied: colorama in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from click>=7.0->uvicorn) (0.4.6)
Requirement already satisfied: annotated-types>=0.6.0 in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from pydantic>=2.9.0->fastapi) (0.8.0)
Requirement already satisfied: pydantic-core==2.46.4 in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from pydantic>=2.9.0->fastapi) (2.46.4)
Requirement already satisfied: anyio<5,>=3.6.2 in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from starlette>=0.46.0->fastapi) (4.14.2)
Requirement already satisfied: idna>=2.8 in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from anyio<5,>=3.6.2->starlette>=0.46.0->fastapi) (3.19)
PS C:\Users\saich\OneDrive\Desktop\HA\skill week5> █
```

python -

### Step 5: Verified FastAPI Installation

```
Requirement already satisfied: idna>=2.8 in C:\Users\saich\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from anyio<5,>=3.6.2->starlette>=0.46.0->fastapi) (3.19)
PS C:\Users\saich\OneDrive\Desktop\HA\skill week5>
PS C:\Users\saich\OneDrive\Desktop\HA\skill week5> python -c "import fastapi; print(fastapi.__version__)"
0.141.1
PS C:\Users\saich\OneDrive\Desktop\HA\skill week5> █
```

## Step 6: Starting the Fast API Server

```
main.py X
main.py > ...
1  from fastapi import FastAPI
2
3  app = FastAPI()
4
5
6  @app.get("/")
7  def home():
8      return {
9          "message": "Welcome to FastAPI"
10     }
```

```
Problems  Output  Debug Console  Terminal  Ports
(.venv) PS C:\Users\saich\OneDrive\Desktop\HA\skill week5> i> python -m pip install fastapi uvicorn
(.venv) PS C:\Users\saich\OneDrive\Desktop\HA\skill week5> i> python -c 'import fastapi; print(fastapi.__version__)'
0.141.1
(.venv) PS C:\Users\saich\OneDrive\Desktop\HA\skill week5> i> python -m uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\Users\duddu\OneDrive\Desktop\DBMS sweek-5']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [8800] using StatReload
INFO: Started server process [29068]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

## Step 7: API Documentation

**FastAPI** 0.1.0 OAS 3.1  
/openapi.json

default

GET / Home

Parameters

No parameters

Execute

Clear

Server response

| Code | Details                               |
|------|---------------------------------------|
| 200  | <div>Response body</div> <pre>{</pre> |

## Step 8: Created First Student API

```
main.py X
main.py > ...
1  from fastapi import FastAPI
2
3  app = FastAPI()
4
5
6  @app.get("/students")
7  def get_students():
8      return [
9          {
10             "id": 101,
11             "name": "Ravi",
12             "department": "CSE"
13          },
14          {
15             "id": 102,
16             "name": "Priya",
17             "department": "CSE"
18          }
19      ]
20
```

### Server response

| Code | Details |
|------|---------|
|------|---------|

|     |  |
|-----|--|
| 200 |  |
|-----|--|

#### Response body

```
[
  {
    "id": 101,
    "name": "Ravi",
    "department": "CSE"
  },
  {
    "id": 102,
    "name": "Priya",
    "department": "CSE"
  }
]
```

#### Response headers

```
content-length: 90
content-type: application/json
date: Wed, 16 Sep 2026 14:21:55 GMT
server: uvicorn
```

Step 9: POST

```
main.py X
main.py > create_student
1  from fastapi import FastAPI
2  from pydantic import BaseModel
3
4  app = FastAPI()
5
6
7  class Student(BaseModel):
8      id: int
9      name: str
10     department: str
11
12
13 @app.post("/students")
14 def create_student(student: Student):
15     return {
16         "message": "Student created",
17         "student": student
18     }
```

FastAPI

0.1.0

OAS 3.1

/openapi.json

default

POST /students Create Student

Server response

| Code | Details                                                                                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 200  | <div>Response body<pre>{  "message": "Student created",  "student": {    "id": 0,    "name": "string",    "department": "string"  }}</pre></div> <div>Response headers<pre>content-length: 86content-type: application/jsondate: Wed, 16 Sep 2026 14:25:42 GMTserver: uvicorn</pre></div> |